MHF4U · GRADE 12 ADVANCED FUNCTIONS

Polynomial Functions

Math Communication Portfolio

Strand C · Polynomial and Rational Functions
Expectations C1, C3, and C4

Name _____

26 October 2026

WHAT THIS PORTFOLIO SHOWS

My understanding of polynomial functions, using the language of the course and work from this unit.

01 C1 Graphs and equations

Degree, end behaviour, finite differences, order of a zero, and the parameters a, k, d, and c.

02 C3 Polynomial equations

Synthetic division, long division, the remainder theorem, the factor theorem, and the rational zero theorem.

03 C4 Polynomial inequalities

Factor tables, interval notation, and a model whose domain has to be restricted.

Learning goals for this unit

These are the polynomial expectations from the MHF4U course. Each artifact is tied to the specific expectation it shows.

C1 Connecting graphs and equations of polynomial functions

Identify and describe key features of polynomial functions, and make connections between the numeric, graphical, and algebraic representations.

C1.1 recognize a polynomial · C1.2 compare representations, including finite differences · C1.3 describe key features · C1.5 factored form and x-intercepts · C1.6 parameters a, k, d, c · C1.8 families of functions · C1.9 even and odd functions

C3 Solving polynomial equations

Solve polynomial equations of degree at most four by factoring, and connect the real roots of an equation to the x-intercepts of the graph.

C3.1 remainder theorem and factor theorem · C3.2 factor polynomials · C3.3 roots and x-intercepts · C3.4 solve equations · C3.7 applications, including unknown coefficients

C4 Solving polynomial inequalities

Explain how the solution of an inequality differs from the solution of an equation, and solve a factorable polynomial inequality.

C4.1 equation versus inequality · C4.2 solve from a graph · C4.3 solve with zeros, a test point or a factor table, and interval notation

SUCCESS CRITERIA

What I am using as evidence of success

I wrote these from the specific expectations. An artifact belongs in the portfolio only if it meets one of them.

C1 Graphs and equations

I can state the degree, the leading coefficient, and the constant term of a polynomial function.

I can give the domain, the range, the end behaviour, and any symmetry, using interval notation.

I can tell a local maximum or minimum from an absolute maximum or minimum.

I can use finite differences to find the degree and the leading coefficient, with $\Delta^n y = a \cdot n!$.

I can use the order of each zero to sketch from factored form and to write a family of functions.

I can describe a, k, d, and c, and apply the mapping $(x, y) \rightarrow (x/k + d, ay + c)$.

I can separate an even or odd degree from an even or odd function.

C3 & C4 Equations and inequalities

I can choose synthetic division for a divisor $x - b$, and long division when the divisor is not linear.

I can write the division statement $P(x) = (\text{divisor})(\text{quotient}) + \text{remainder}$, with the remainder of smaller degree.

I can use the remainder theorem, including a divisor $ax - b$, where the remainder is $P(b/a)$.

I can use the factor theorem and the rational zero theorem to factor a cubic or a quartic.

I can state the roots of an equation and, separately, the x-intercepts of the graph.

I can solve a polynomial inequality with a factor table and matching interval notation.

I can restrict an answer so that it makes sense in the context of the problem.

What the graph is telling me before I plot points

End behaviour is decided first

The degree tells me whether the ends behave in the same way or in opposite ways. The leading coefficient tells me which quadrants. A positive odd degree extends from Quadrant 3 to Quadrant 1. A negative odd degree extends from Quadrant 2 to Quadrant 4. A positive even degree extends from Quadrant 2 to Quadrant 1. A negative even degree extends from Quadrant 3 to Quadrant 4.

Degree limits the graph. It does not fill it in.

A polynomial of degree n has at most n distinct zeros and at most n − 1 turning points. An odd degree must have at least one real zero, because the ends go to opposite infinities, so the graph has to cross the x-axis. An even degree can have no x-intercept at all. A particular graph can have fewer turning points than the maximum.

Even degree is not an even function.

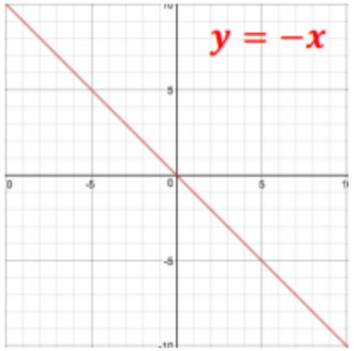
An even function satisfies f(−x) = f(x) and has line symmetry about the y-axis. An odd function satisfies f(−x) = −f(x) and has point symmetry about the origin. Every exponent has to agree, and an odd function cannot have a constant term. $f(x) = x^3 + 1$ is degree 3, but it is neither even nor odd.

Key features of y = −x

Example 4: For each of the following functions

i) State the domain and range
ii) Describe the end behavior
iii) Identify any symmetry

a)



i) Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$

ii) As $x \to -\infty, y \to \infty$ and as $x \to \infty, y \to -\infty$
The graph extends from quadrant 2 to 4

iii) Point symmetry about the origin (0, 0)

What this work shows

y = −x is a power function, $f(x) = ax^n$, with a = −1 and n = 1. The domain and the range are both (−∞, ∞).

End behaviour: as x → −∞, y → ∞, and as x → ∞, y → −∞. The graph extends from Quadrant 2 to Quadrant 4. That is the negative odd-degree case.

It has point symmetry about the origin, so it is an odd function. I can also check the rule: f(−x) = −(−x) = x = −f(x).

The degree being odd is not what makes it an odd function. f(x) = −x + 1 is still degree 1, but f(−x) = x + 1, which is neither f(x) nor −f(x). The constant term breaks the point symmetry.

Even and odd: the degree compared with the function

A polynomial can have an even or odd degree without being an even or odd function. The function test is algebraic.

	Even function	Odd function	Neither
Symmetry	Line symmetry about the y-axis	Point symmetry about (0, 0)	Neither of those symmetries
Rule	$f(-x) = f(x)$	$f(-x) = -f(x)$	Neither equation holds
Exponents	Every exponent is even	Every exponent is odd	The exponents are mixed
Constant term	A constant term is allowed	The constant term must be 0	A constant term is allowed
Example	$f(x) = x^4 - x^2$	$f(x) = -2x^3 + 5x$	$f(x) = x^3 + 1$
End behaviour	Same on both ends	Opposite ends	Depends on the degree and on a

On the characteristics quiz, the statement “all odd-degree polynomial functions are odd functions” is false. $f(x) = x^3 + 1$ has odd degree, but $f(-x) = -x^3 + 1$, and $-f(x) = -x^3 - 1$. Those are not equal. The same quiz item is why I keep “degree” and “function” as two different words.

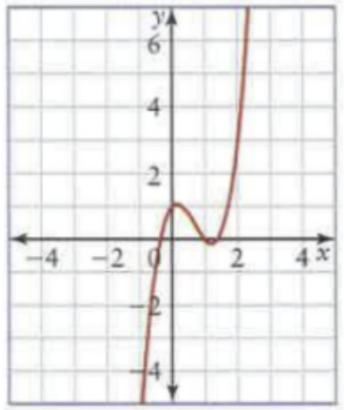
Reading a cubic from its equation

$f(x) = 2x^3 - 4x^2 + x + 1$

a) The function is cubic with a positive leading coefficient. The graph extends from Q3 to Q1. The y-intercept is 1. Graph iv) corresponds to this equation.

There are 3 x-intercepts and the degree is 3. The function has one local max and one local min, which is a total of two turning points, which is one less than the degree. There is no absolute max or min point.

iv)



f(x) = 2x³ − 4x² + x + 1

Degree 3, leading coefficient 2 > 0, so the graph extends from Quadrant 3 to Quadrant 1. The constant term is the y-intercept, so the y-intercept is 1. That matched graph (iv).

The graph has 3 x-intercepts, the most a cubic can have, and 2 turning points, which is n − 1: one local maximum and one local minimum. There is no absolute maximum and no absolute minimum, because an odd-degree polynomial is unbounded in both directions.

I checked the zeros afterwards. f(1) = 0, so x − 1 is a factor. The quotient 2x² − 2x − 1 has two more real zeros, since its discriminant is 4 + 8 = 12. Three real zeros agrees with the graph.

Finite differences: the table I left blank, completed

x	y	First	Second	Third
−3	−36	—	—	—
−2	−12	24	—	—
−1	−2	10	−14	—
0	0	2	−8	6
1	0	0	−2	6
2	4	4	4	6
3	18	14	10	6
4	48	30	16	6

What the constant row tells me

The third finite differences are the first ones that are constant, so the degree is 3.

They are positive, so the leading coefficient is positive. The value comes from $\Delta^n y = a \cdot n!$.

$\Delta^3 y = a \cdot 3! = 6a$

$6 = 6a$, so $a = 1$.

The polynomial that fits the table is $y = x^3 - x^2$. It has leading coefficient 1, and it matches every value, including $y(-1) = -2$.

This is the same rule as the quiz item where the fourth finite differences were −48: $\Delta^4 y = a \cdot 4! = 24a$, so $a = -2$.

Order of a zero, and where the graph changes sign

$f(x) = (x + 1)^2(x + 2)(x - 1)$ degree 4, leading coefficient positive, y-intercept (0, −2)

	x = −2	x = −1	x = 1
Factor	$(x + 2)$	$(x + 1)^2$	$(x - 1)$
Order	1	2	1
Shape at the x-axis	Crosses, linear shape	Bounces, parabolic shape	Crosses, linear shape
Sign of f(x)	Changes	Does not change	Changes

SIGN OF f(x)

Interval	Sign
(−∞, −2)	+
(−2, −1)	−
(−1, 1)	−
(1, ∞)	+

The course uses order for the exponent on a factor. A zero of odd order changes the sign: the graph crosses the x-axis. Order 1 crosses in a linear shape. Order 3 flattens and crosses in an S-shape. A zero of even order does not change the sign. Order 2 bounces off the x-axis.

On my sign chart the sign stays negative on both sides of x = −1. That is the order 2 zero. The two order 1 zeros are where the sign changes. These zeros do not give only one equation. The family is $f(x) = k(x + 1)^2(x + 2)(x - 1)$, $k \neq 0$. One other point determines k. That is C1.8.

Getting g(x) into the form $a[k(x - d)]^n + c$

The parent is $f(x) = x^4$. The given equation is $g(x) = -((1/3)x + 2)^4 - 1$. I factored k out before I named d:

$g(x) = -[(1/3)(x + 6)]^4 - 1$

a = −1

Reflection in the x-axis. |a| = 1, so there is no vertical stretch or compression.

k = 1/3

Horizontal stretch by a factor of 3. k is positive, so there is no reflection in the y-axis.

d = −6

Translation 6 units left. The +2 in the original equation was part of k, not the value of d.

c = −1

Translation 1 unit down. Stretches and reflections, from a and k, come before translations.

Mapping rule

$(x, y) \rightarrow (x/k + d, ay + c) = (3x - 6, -y - 1)$

The point (2, 16) on the parent maps to (0, −17). That matches the table in my notes. Domain: (−∞, ∞). The parent range [0, ∞) is reflected to (−∞, 0] and then moved down 1, so the range is (−∞, −1]. The absolute maximum point is (−6, −1).

End behaviour: even degree, negative leading coefficient, Quadrant 3 to Quadrant 4. The parent has line symmetry about the y-axis. After the translation, the line of symmetry is x = −6. g is not an even function.

A cubic image point, with a and k both different from 1

Parent $f(x) = x^3$. $g(x) = -2[3(x - 1)]^3 - 5$

The point (2, 8) lies on the parent. I mapped it after every transformation.

1 Name the parameters

$a = -2$, $k = 3$, $d = 1$, $c = -5$

2 Write the mapping

$(x, y) \rightarrow (x/3 + 1,\ -2y - 5)$

3 Substitute (2, 8)

$(2/3 + 1,\ -16 - 5) = (5/3,\ -21)$

Key features of g

Point of inflection, the centre of point symmetry: (d, c) = (1, −5).

Domain (−∞, ∞). Range (−∞, ∞), because the degree is odd.

End behaviour: odd degree and $a < 0$, so the graph extends from Quadrant 2 to Quadrant 4.

One valid order

Horizontal compression by a factor of 1/3, vertical stretch by a factor of 2, and a reflection in the x-axis.

Then a translation 1 unit right and 5 units down.

Stretches and reflections come before translations. Translating first would move the wrong point.

Division is how an equation becomes factors

Remainder theorem

When P(x) is divided by x − b, the remainder is P(b). When the divisor is ax − b, with a ≠ 0, the remainder is P(b/a). I do not have to divide if all I need is the remainder.

Factor theorem

x − b is a factor of P(x) if and only if P(b) = 0. For a divisor ax − b, the test is P(b/a) = 0. A zero remainder means the divisor is a factor, so it can be used to reduce the degree.

Rational zero theorem

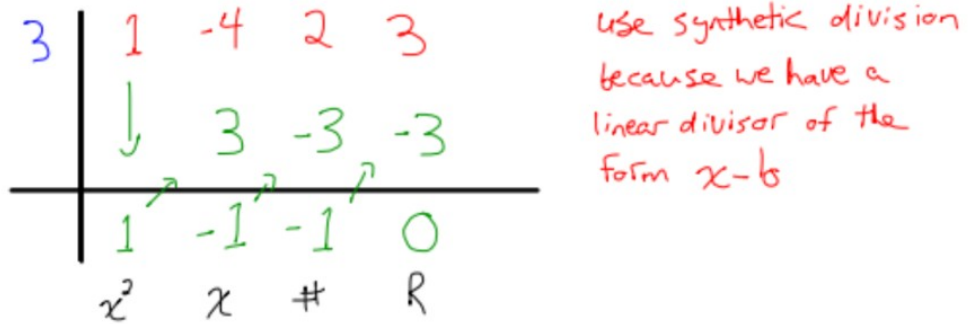
If a rational number p/q, in lowest terms, is a zero, then p is a factor of the constant term and q is a factor of the leading coefficient. The integral zero theorem is the case where the leading coefficient is 1, so q = 1.

Division algorithm

P(x) = (divisor)(quotient) + remainder, and the degree of the remainder is less than the degree of the divisor. A linear divisor leaves a constant remainder. A quadratic divisor leaves a remainder of the form ax + b.

The divisor decides the method

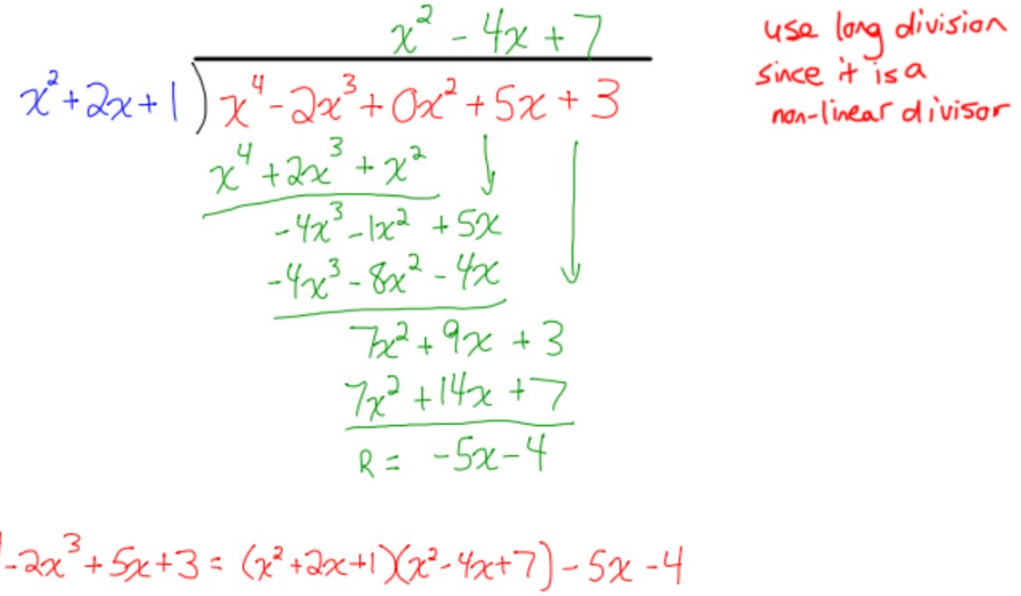
a) Divide $x^3 - 4x^2 + 2x + 3$ by $x - 3$



Synthetic division, because the divisor is x − b.

$x^3 - 4x^2 + 2x + 3 = (x - 3)(x^2 - x - 1)$. The remainder is 0, so both factors divide the dividend. In quotient form, the restriction is $x \neq 3$.

c) Divide $x^4 - 2x^3 + 5x + 3$ by $x^2 + 2x + 1$



Long division, because the divisor is not linear.

I inserted $0x^2$ so no power was missing. The remainder is $-5x - 4$. A degree 2 divisor can leave a linear remainder. Synthetic division does not apply here.

A root is an x-value. An x-intercept is a point.

Factor $P(x) = 2x^3 + x^2 - 7x - 6$

Possible values of b, factors of the constant term −6: ±1, ±2, ±3, ±6.

Possible values of a, factors of the leading coefficient 2: ±1, ±2.

Possible rational zeros b/a: ±1, ±1/2, ±2, ±3, ±3/2, ±6.

P(−1) = 0, so x + 1 is a factor. Synthetic division gives the quotient $2x^2 - x - 6$.

$2x^2 - x - 6 = (2x + 3)(x - 2)$, by grouping.

$P(x) = (x + 1)(x - 2)(2x + 3)$

Zeros: x = −1, x = 2, x = −3/2.

x-intercepts: (−1, 0), (2, 0), (−3/2, 0).

The equation I solved

$2x^3 + 3x^2 - 11x - 6 = 0$

By the rational zero theorem I tested x = 2. f(2) = 0, so x − 2 is a factor. The quotient is $2x^2 + 7x + 3 = (2x + 1)(x + 3)$.

$(x - 2)(2x + 1)(x + 3) = 0$

The solutions of the equation, the roots, are x = 2, x = −1/2, and x = −3.

In my lesson note I wrote the solutions as the points (2, 0), (−1/2, 0), and (−3, 0). Those points are the x-intercepts of y = f(x). C3.3 asks for the connection between the two, and it also asks me to keep the names separate. The equation is solved by the three x-values.

When the remainder is not a constant

Find the remainder when $P(x) = x^3 + 3x^2 - x - 2$ is divided by $(x + 3)(x + 5)$.

Why ax + b

The divisor has degree 2. The division algorithm says the remainder has degree less than 2, so $R(x) = ax + b$. It is not a single number.

Use the zeros

$P(x) = (x + 3)(x + 5)Q(x) + ax + b$. At a zero of the divisor, the product is 0, so $P(\text{zero}) = R(\text{zero})$.

Two equations

$P(-3) = 1$ gives $-3a + b = 1$. $P(-5) = -47$ gives $-5a + b = -47$. Subtracting gives $2a = 48$.

The remainder

$a = 24$ and $b = 73$. The remainder is $24x + 73$. I can check: $R(-3) = -72 + 73 = 1$, and $R(-5) = -120 + 73 = -47$.

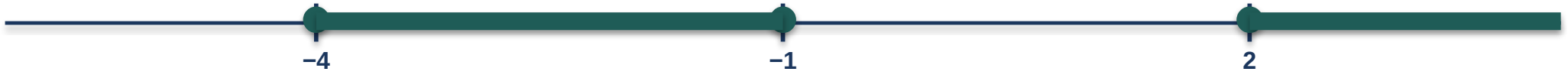
The same idea, with an unknown polynomial

If $f(x)$ leaves a remainder of 3 when it is divided by $x + 2$, then $f(-2) = 3$. For $g(x) = f(x) + 4x + 7$, the remainder on division by $x + 2$ is $g(-2) = 3 + 4(-2) + 7 = 2$. I never need an expression for $f(x)$. The remainder theorem is a statement about the value of the function, not about finding every coefficient.

Solve $-2x^3 - 6x^2 + 12x \leq -16$

SOLUTION ON THE NUMBER LINE

$[-4, -1] \cup [2, \infty)$



Algebra

Move every term to one side. The symbol stays ≤ until I multiply or divide by a negative.

$-2x^3 - 6x^2 + 12x + 16 \leq 0$

$-2(x^3 + 3x^2 - 6x - 8) \leq 0$

Divide by −2 and reverse the inequality.

$(x + 4)(x + 1)(x - 2) \geq 0$

The product is positive or zero on [−4, −1] and on [2, ∞). At each zero the polynomial equals 0, and 0 satisfies ≥, so the brackets are closed.

Brackets, and the list of possible zeros

The solution I wrote, $-4 \leq x \leq -1$ or $x \geq 2$, matches the interval notation. The brackets are closed because the inequality is inclusive. That is the difference C4.1 asks for: the equation has three solutions, and the inequality has two intervals.

In the list of candidates I wrote ±1, ±2, and ±3 as factors of −8. The integral zero theorem only allows integer factors of the constant term. 3 does not divide 8. The full list is ±1, ±2, ±4, ±8. I still found the factor x + 1, so the zeros did not change, but the list of possible rational zeros was not complete.

A volume model, with a restriction from the context

$V(y) = 2y^3 + y^2 - 27y - 36$ gives the volume, in cm^3. The height is $h(y) = y + 3$.

Divide by the height

The volume is length times width times height, so the quotient is the area of the base. Synthetic division with root −3 gives quotient $2y^2 - 5y - 12$ and remainder 0. The factor theorem agrees: a given linear factor must make the remainder 0.

Factor the base

$2y^2 - 5y - 12 = (2y + 3)(y - 4)$. So one possible set of dimensions, in centimetres, is $y + 3$, $2y + 3$, and $y - 4$. Length and width can be swapped. The algebra does not decide which is which.

Restrict the domain

A dimension of a box has to be positive. $y + 3 > 0$ and $2y + 3 > 0$ are both true once $y - 4 > 0$. The model describes a possible box only when $y > 4$. That restriction is part of the solution, not an extra comment.

Division statement

$2y^3 + y^2 - 27y - 36 = (y + 3)(2y + 3)(y - 4)$, for $y > 4$.

ACROSS C1, C3, AND C4

The graph, the equation, and the inequality are one polynomial

01

Same zeros, three questions

f(x) = 0 asks for the roots. The graph shows those roots as x-intercepts, and the order of each zero tells me whether the graph crosses or turns. f(x) ≥ 0 asks which intervals between those zeros make the product positive or zero. A factor table is the sign chart I already used when I sketched from factored form.

02

The algebra is the bridge

Synthetic division and the factor theorem turn an expanded polynomial into factors. The division algorithm tells me what degree of remainder to expect, including a linear remainder when the divisor is quadratic. Finite differences and the leading coefficient tell me the end behaviour when I start from a table instead of an equation.

03

The precise words are the mathematics

A root is an x-value. An x-intercept is a point. An odd degree does not make an odd function. k has to be factored into k(x − d) before I name the translation. Square brackets belong to an inclusive inequality, because the zeros satisfy ≥ or ≤ and do not satisfy a strict inequality. Those are the places my own notes had to get more careful.